```
import { useState, useEffect, useRef } from "react";

const FOODS = {
  "Weight Loss": {
    breakfast: [
      { name: "Greek Yogurt Parfait", cal: 320, protein: 22, carbs: 38, fat: 8, e
      { name: "Veggie Omelette", cal: 280, protein: 18, carbs: 6, fat: 18, emoji:
      { name: "Overnight Oats", cal: 350, protein: 14, carbs: 52, fat: 9, emoji:
    ],
    lunch: [
      { name: "Grilled Chicken Salad", cal: 420, protein: 38, carbs: 22, fat: 14,
      { name: "Turkey Lettuce Wraps", cal: 360, protein: 32, carbs: 18, fat: 12,
      { name: "Quinoa Buddha Bowl", cal: 480, protein: 20, carbs: 58, fat: 16, em
    ],
    dinner: [
      { name: "Baked Salmon + Broccoli", cal: 520, protein: 44, carbs: 18, fat: 2
      { name: "Zucchini Noodles + Turkey", cal: 440, protein: 36, carbs: 22, fat:
      { name: "Cauliflower Fried Rice", cal: 390, protein: 28, carbs: 32, fat: 14
    ],
    snacks: [
      { name: "Apple + Almond Butter", cal: 180, protein: 4, carbs: 22, fat: 9, e
      { name: "Hummus + Veggie Sticks", cal: 150, protein: 6, carbs: 18, fat: 7,
    ],
  },
  "Muscle Gain": {
    breakfast: [
      { name: "Protein Pancakes", cal: 520, protein: 38, carbs: 52, fat: 12, emoj
      { name: "Egg & Sweet Potato Bowl", cal: 580, protein: 32, carbs: 62, fat: 1
      { name: "Smoothie Bowl + Granola", cal: 620, protein: 28, carbs: 78, fat: 1
    ],
    lunch: [
      { name: "Chicken Rice Power Bowl", cal: 680, protein: 52, carbs: 72, fat: 1
      { name: "Tuna Pasta Salad", cal: 640, protein: 44, carbs: 66, fat: 16, emoj
      { name: "Lean Beef Stir Fry", cal: 720, protein: 48, carbs: 62, fat: 22, em
    ],
    dinner: [
      { name: "Steak + Sweet Potato", cal: 780, protein: 56, carbs: 58, fat: 26,
      { name: "Salmon + Brown Rice", cal: 720, protein: 48, carbs: 64, fat: 24, e
      { name: "Chicken Curry + Quinoa", cal: 760, protein: 52, carbs: 68, fat: 22
    ],
    snacks: [
      { name: "Cottage Cheese + Berries", cal: 220, protein: 18, carbs: 22, fat:
      { name: "Protein Shake + Banana", cal: 280, protein: 26, carbs: 36, fat: 5,
    ],
```

```
  },
  "Maintenance": {
    breakfast: [
      { name: "Avocado Toast + Egg", cal: 420, protein: 18, carbs: 42, fat: 22, e
      { name: "Fruit Salad + Yogurt", cal: 380, protein: 14, carbs: 58, fat: 8, e
      { name: "Whole Grain Porridge", cal: 400, protein: 12, carbs: 68, fat: 9, e
    ],
    lunch: [
      { name: "Mediterranean Wrap", cal: 560, protein: 28, carbs: 54, fat: 22, em
      { name: "Lentil Soup + Bread", cal: 480, protein: 22, carbs: 72, fat: 10, e
      { name: "Grilled Veggie Panini", cal: 520, protein: 20, carbs: 62, fat: 18,
    ],
    dinner: [
      { name: "Pasta Primavera", cal: 580, protein: 22, carbs: 82, fat: 16, emoji
      { name: "Grilled Chicken Tacos", cal: 620, protein: 36, carbs: 64, fat: 20,
      { name: "Vegetable Curry + Rice", cal: 560, protein: 18, carbs: 78, fat: 16
    ],
    snacks: [
      { name: "Mixed Nuts + Dried Fruit", cal: 200, protein: 6, carbs: 22, fat: 1
      { name: "Dark Chocolate + Berries", cal: 180, protein: 3, carbs: 26, fat: 9
    ],
  },
  "Diabetes Management": {
    breakfast: [
      { name: "Egg White Omelette", cal: 240, protein: 20, carbs: 8, fat: 12, emo
      { name: "Chia Seed Pudding", cal: 280, protein: 12, carbs: 28, fat: 14, emo
      { name: "Spinach Smoothie", cal: 260, protein: 14, carbs: 30, fat: 8, emoji
    ],
    lunch: [
      { name: "Grilled Fish Salad", cal: 380, protein: 36, carbs: 18, fat: 14, em
      { name: "Chickpea & Veg Soup", cal: 340, protein: 18, carbs: 42, fat: 8, em
      { name: "Turkey & Avocado Bowl", cal: 420, protein: 32, carbs: 28, fat: 20,
    ],
    dinner: [
      { name: "Steamed Cod + Broccoli", cal: 380, protein: 42, carbs: 16, fat: 12
      { name: "Tofu Stir Fry + Cauliflower Rice", cal: 360, protein: 24, carbs: 2
      { name: "Baked Chicken + Asparagus", cal: 420, protein: 44, carbs: 14, fat:
    ],
    snacks: [
      { name: "Celery + Almond Butter", cal: 130, protein: 4, carbs: 8, fat: 10,
      { name: "Hard Boiled Eggs", cal: 140, protein: 12, carbs: 1, fat: 10, emoji
    ],
  },
};

const ML_INSIGHTS = [
  { icon: "🧠", title: "Collaborative Filtering", desc: "Matched with 2,847 simil
```

```
  { icon: "📊", title: "Content-Based ML", desc: "Optimized for your macronutrien
  { icon: "🔄", title: "Reinforcement Learning", desc: "Adapting based on your fe
  { icon: "👁️", title: "Computer Vision", desc: "Food recognition model: ResNet-5
];

const NUTRIENT_COLORS = { protein: "#e74c3c", carbs: "#f39c12", fat: "#3498db" };

function NutrientBar({ label, value, max, color }) {
  const pct = Math.min((value / max) * 100, 100);
  return (
    <div style={{ marginBottom: 10 }}>
      <div style={{ display: "flex", justifyContent: "space-between", marginBotto
        <span style={{ fontWeight: 600, textTransform: "uppercase", letterSpacing
        <span style={{ color }}>{value}g / {max}g</span>
      </div>
      <div style={{ background: "#eee", borderRadius: 6, height: 8, overflow: "hi
        <div style={{
          width: `${pct}%`, height: "100%", background: color,
          borderRadius: 6, transition: "width 0.8s cubic-bezier(.4,0,.2,1)"
        }} />
      </div>
    </div>
  );
}

function FoodCard({ item, meal, onRate }) {
  const [rated, setRated] = useState(null);
  const handleRate = (v) => { setRated(v); onRate && onRate(v); };
  return (
    <div style={{
      background: "white", borderRadius: 16, padding: "16px 18px", marginBottom:
      border: "1px solid #e8f5e9", boxShadow: "0 2px 12px #1a6b3a11",
      transition: "transform 0.2s, box-shadow 0.2s",
    }}
      onMouseEnter={e => { e.currentTarget.style.transform = "translateY(-2px)";
      onMouseLeave={e => { e.currentTarget.style.transform = ""; e.currentTarget.
    >
      <div style={{ display: "flex", alignItems: "flex-start", gap: 14 }}>
        <div style={{
          fontSize: 36, width: 54, height: 54, display: "flex", alignItems: "cent
          justifyContent: "center", background: "#e8f5e9", borderRadius: 14, flex
        }}>{item.emoji}</div>
        <div style={{ flex: 1 }}>
          <div style={{ display: "flex", justifyContent: "space-between", alignIt
            <div>
              <div style={{ fontWeight: 700, fontSize: 15, color: "#1a2e1a" }}>{i
              <div style={{ fontSize: 12, color: "#888", marginTop: 2 }}>⏱ {item
```

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            </div>
            <div style={{
              background: "#e8f5e9", color: "#1a6b3a", fontWeight: 800,
              fontSize: 14, borderRadius: 10, padding: "4px 10px"
            }}>{item.cal} kcal</div>
          </div>
          <div style={{ display: "flex", gap: 12, marginTop: 10 }}>
            {[["P", item.protein, "#e74c3c"], ["C", item.carbs, "#f39c12"], ["F",
              <div key={l} style={{
                background: c + "15", color: c, borderRadius: 8,
                padding: "3px 10px", fontSize: 12, fontWeight: 700
              }}>{l}: {v}g</div>
            ))}
          </div>
        </div>
      </div>
      <div style={{ display: "flex", gap: 8, marginTop: 12, borderTop: "1px solid
        <span style={{ fontSize: 11, color: "#888", marginRight: "auto", alignSel
          Rate this recommendation:
        </span>
        {["👍", "👎"].map((em, i) => (
          <button key={em} onClick={() => handleRate(i === 0 ? 1 : -1)} style={{
            border: "none", background: rated === (i === 0 ? 1 : -1) ? (i === 0 ?
            borderRadius: 8, padding: "4px 14px", cursor: "pointer", fontSize: 16
            transition: "all 0.2s", outline: "none"
          }}>{em}</button>
        ))}
      </div>
    </div>
  );
}

function MealSection({ title, icon, items, onRate }) {
  const [open, setOpen] = useState(true);
  return (
    <div style={{ marginBottom: 24 }}>
      <button onClick={() => setOpen(o => !o)} style={{
        display: "flex", alignItems: "center", gap: 10, background: "none",
        border: "none", cursor: "pointer", width: "100%", marginBottom: open ? 14
      }}>
        <span style={{
          fontSize: 22, width: 40, height: 40, background: "#e8f5e9",
          borderRadius: 12, display: "flex", alignItems: "center", justifyContent
        }}>{icon}</span>
        <span style={{ fontWeight: 800, fontSize: 16, color: "#1a2e1a", letterSpa
        <span style={{ marginLeft: "auto", fontSize: 18, color: "#1a6b3a", transi
      </button>
```

```
      {open && items.map((item, i) => (
        <FoodCard key={i} item={item} meal={title} onRate={v => onRate(item.name,
      ))}
    </div>
  );
}

function AIInsightsPanel() {
  const [active, setActive] = useState(0);
  useEffect(() => {
    const t = setInterval(() => setActive(a => (a + 1) % ML_INSIGHTS.length), 220
    return () => clearInterval(t);
  }, []);
  return (
    <div style={{ background: "#f0faf2", borderRadius: 18, padding: "18px 20px",
      <div style={{ fontWeight: 800, fontSize: 13, color: "#1a6b3a", letterSpacin
        🤖 Active AI Models
      </div>
      <div style={{ display: "grid", gridTemplateColumns: "1fr 1fr", gap: 10 }}>
        {ML_INSIGHTS.map((item, i) => (
          <div key={i} style={{
            background: active === i ? item.color : "white",
            color: active === i ? "white" : "#333",
            borderRadius: 12, padding: "10px 14px", transition: "all 0.5s cubic-b
            border: `1px solid ${active === i ? item.color : "#e0e0e0"}`,
          }}>
            <div style={{ fontSize: 20, marginBottom: 4 }}>{item.icon}</div>
            <div style={{ fontWeight: 700, fontSize: 12 }}>{item.title}</div>
            <div style={{ fontSize: 11, opacity: 0.85, marginTop: 2 }}>{item.desc
          </div>
        ))}
      </div>
    </div>
  );
}

export default function App() {
  const [step, setStep] = useState("onboard");
  const [profile, setProfile] = useState({
    name: "", age: "", weight: "", height: "", goal: "", diet: "", activity: ""
  });
  const [plan, setPlan] = useState(null);
  const [ratings, setRatings] = useState({});
  const [loading, setLoading] = useState(false);
  const [aiResponse, setAiResponse] = useState("");
  const [chatInput, setChatInput] = useState("");
  const [chatHistory, setChatHistory] = useState([]);
```

```
const [chatLoading, setChatLoading] = useState(false);
const chatRef = useRef(null);

const goals = ["Weight Loss", "Muscle Gain", "Maintenance", "Diabetes Managemen
const diets = ["Omnivore", "Vegetarian", "Vegan", "Keto", "Mediterranean"];
const activities = ["Sedentary", "Light", "Moderate", "Active", "Very Active"];

const bmi = profile.weight && profile.height
  ? (profile.weight / ((profile.height / 100) ** 2)).toFixed(1) : null;

const activityMultipliers = { Sedentary: 1.2, Light: 1.375, Moderate: 1.55, Act
const bmr = profile.age && profile.weight && profile.height
  ? Math.round((10 * profile.weight) + (6.25 * profile.height) - (5 * profile.a
const tdee = bmr ? Math.round(bmr * (activityMultipliers[profile.activity] || 1
const targetCal = goal => {
  if (goal === "Weight Loss") return Math.round(tdee * 0.8);
  if (goal === "Muscle Gain") return Math.round(tdee * 1.1);
  return tdee;
};

const generatePlan = () => {
  setLoading(true);
  setTimeout(() => {
    const foodSet = FOODS[profile.goal] || FOODS["Maintenance"];
    const pick = arr => arr[Math.floor(Math.random() * arr.length)];
    setPlan({
      breakfast: pick(foodSet.breakfast),
      lunch: pick(foodSet.lunch),
      dinner: pick(foodSet.dinner),
      snack1: pick(foodSet.snacks),
      snack2: pick(foodSet.snacks.filter(s => s.name !== pick(foodSet.snacks).n
    });
    setLoading(false);
    setStep("plan");
  }, 2000);
};

const totalNutrients = plan ? {
  cal: [plan.breakfast, plan.lunch, plan.dinner, plan.snack1, plan.snack2].redu
  protein: [plan.breakfast, plan.lunch, plan.dinner, plan.snack1, plan.snack2].
  carbs: [plan.breakfast, plan.lunch, plan.dinner, plan.snack1, plan.snack2].re
  fat: [plan.breakfast, plan.lunch, plan.dinner, plan.snack1, plan.snack2].redu
} : null;

const sendChat = async () => {
  if (!chatInput.trim() || chatLoading) return;
  const userMsg = chatInput;
```

```
    setChatInput("");
    const newHistory = [...chatHistory, { role: "user", content: userMsg }];
    setChatHistory(newHistory);
    setChatLoading(true);
    try {
      const systemPrompt = `You are NutriAI, an expert AI dietitian assistant emb
The user profile: Name=${profile.name || "User"}, Age=${profile.age}, Weight=${pr
Current meal plan: Breakfast: ${plan?.breakfast?.name || "none"}, Lunch: ${plan?.
Be helpful, scientific, warm, and concise (3-4 sentences max). Never give medical
      const response = await fetch("https://api.anthropic.com/v1/messages", {
        method: "POST",
        headers: { "Content-Type": "application/json" },
        body: JSON.stringify({
          model: "claude-sonnet-4-20250514",
          max_tokens: 1000,
          system: systemPrompt,
          messages: newHistory.map(m => ({ role: m.role, content: m.content }))
        })
      });
      const data = await response.json();
      const reply = data.content?.map(c => c.text || "").join("") || "Sorry, I co
      setChatHistory(h => [...h, { role: "assistant", content: reply }]);
    } catch {
      setChatHistory(h => [...h, { role: "assistant", content: "Sorry, I'm having
    }
    setChatLoading(false);
    setTimeout(() => chatRef.current?.scrollTo({ top: chatRef.current.scrollHeigh
  };

  const handleRating = (food, val) => {
    setRatings(r => ({ ...r, [food]: val }));
  };

  const filled = profile.name && profile.age && profile.weight && profile.height

  // ─── ONBOARDING ──────────────────────────────────────────────
  if (step === "onboard") return (
    <div style={{
      minHeight: "100vh", background: "linear-gradient(135deg, #e8f5e9 0%, #f1f8e
      display: "flex", alignItems: "center", justifyContent: "center", fontFamily
    }}>
      <div style={{ width: "100%", maxWidth: 560 }}>
        <div style={{ textAlign: "center", marginBottom: 32 }}>
          <div style={{ fontSize: 64, marginBottom: 12 }}>🥗</div>
          <h1 style={{ margin: 0, fontSize: 28, fontWeight: 900, color: "#1a2e1a"
            NutriAI
          </h1>
```

```
  <p style={{ color: "#4a7a4a", margin: "6px 0 0", fontSize: 15 }}>
    AI-Powered Diet Recommendation System
  </p>
  <div style={{ display: "flex", justifyContent: "center", gap: 10, margi
    {["🤖 ML-Powered", "📊 Personalized", "🔬 Evidence-Based"].map(tag =>
      <span key={tag} style={{
        background: "#e8f5e9", color: "#1a6b3a", borderRadius: 20,
        padding: "4px 14px", fontSize: 12, fontWeight: 600
      }}>{tag}</span>
    ))}
  </div>
</div>

<div style={{
  background: "white", borderRadius: 24, padding: "32px 36px",
  boxShadow: "0 12px 60px #1a6b3a22", border: "1px solid #e8f5e9"
}}>
  <h2 style={{ margin: "0 0 24px", fontSize: 18, color: "#1a2e1a", fontWe
    Build Your Profile
  </h2>

  <div style={{ display: "grid", gridTemplateColumns: "1fr 1fr", gap: 14,
    {[["Your Name", "name", "text", "e.g. Alex"], ["Age (years)", "age",
      ["Weight (kg)", "weight", "number", "e.g. 72"], ["Height (cm)", "he
    ].map(([label, key, type, ph]) => (
      <div key={key}>
        <label style={{ fontSize: 11, fontWeight: 700, color: "#666", let
        <input
          type={type} placeholder={ph} value={profile[key]}
          onChange={e => setProfile(p => ({ ...p, [key]: e.target.value }
          style={{
            width: "100%", borderRadius: 10, border: "1.5px solid #e0e0e0
            padding: "10px 12px", fontSize: 14, marginTop: 6, outline: "n
            boxSizing: "border-box", transition: "border 0.2s",
            fontFamily: "inherit"
          }}
          onFocus={e => e.target.style.borderColor = "#1a6b3a"}
          onBlur={e => e.target.style.borderColor = "#e0e0e0"}
        />
      </div>
    ))}
  </div>

  {[["Health Goal", "goal", goals], ["Dietary Type", "diet", diets], ["Ac
    <div key={key} style={{ marginBottom: 16 }}>
      <label style={{ fontSize: 11, fontWeight: 700, color: "#666", lette
      <div style={{ display: "flex", flexWrap: "wrap", gap: 8, marginTop:
```

```
              {opts.map(opt => (
                <button key={opt} onClick={() => setProfile(p => ({ ...p, [key]
                  style={{
                    padding: "8px 14px", borderRadius: 10, border: "1.5px solid
                    borderColor: profile[key] === opt ? "#1a6b3a" : "#e0e0e0",
                    background: profile[key] === opt ? "#1a6b3a" : "white",
                    color: profile[key] === opt ? "white" : "#555",
                    fontSize: 13, fontWeight: 600, cursor: "pointer", transitio
                  }}>{opt}</button>
              ))}
            </div>
          </div>
        ))}

        {bmi && (
          <div style={{
            background: "#f0faf2", borderRadius: 12, padding: "12px 16px",
            marginBottom: 20, border: "1px solid #c8e6c9", display: "flex", gap
          }}>
            <div><div style={{ fontSize: 11, color: "#666", fontWeight: 700 }}>
              <div style={{ fontSize: 20, fontWeight: 900, color: "#1a6b3a" }}>
            <div><div style={{ fontSize: 11, color: "#666", fontWeight: 700 }}>
              <div style={{ fontSize: 20, fontWeight: 900, color: "#1a6b3a" }}>
            <div><div style={{ fontSize: 11, color: "#666", fontWeight: 700 }}>
              <div style={{ fontSize: 20, fontWeight: 900, color: "#1a6b3a" }}>
          </div>
        )}

        <button onClick={generatePlan} disabled={!filled}
          style={{
            width: "100%", padding: "14px", borderRadius: 14, border: "none",
            background: filled ? "linear-gradient(135deg, #1a6b3a, #2e7d32)" :
            color: "white", fontSize: 16, fontWeight: 800, cursor: filled ? "po
            letterSpacing: 0.5, transition: "all 0.2s",
            boxShadow: filled ? "0 6px 20px #1a6b3a44" : "none"
          }}>
          🤖 Generate My AI Meal Plan
        </button>
      </div>
    </div>
  </div>
);

// ─── LOADING ──────────────────────────────────────────────
if (loading) return (
  <div style={{
    minHeight: "100vh", background: "linear-gradient(135deg, #e8f5e9, #f1f8e9)"
```

```
    display: "flex", alignItems: "center", justifyContent: "center", fontFamily
  }}>
    <div style={{ textAlign: "center" }}>
      <div style={{ fontSize: 64, animation: "spin 2s linear infinite", display
      <style>{`@keyframes spin { from { transform: rotate(0deg) } to { transfor
      <h2 style={{ color: "#1a6b3a", marginTop: 24, fontWeight: 900 }}>AI is cr
      {["Running collaborative filtering...", "Analyzing 50,000+ meals...", "Op
        <div key={i} style={{
          color: "#4a7a4a", fontSize: 14, padding: "6px 0",
          animation: `fadeIn 0.5s ${i * 0.4}s both`
        }}>
          <style>{`@keyframes fadeIn { from { opacity: 0; transform: translateY
          ✓ {msg}
        </div>
      ))}
    </div>
  </div>
);

// ─── PLAN VIEW ──────────────────────────────────────────────
if (step === "plan" && plan) return (
  <div style={{
    minHeight: "100vh", background: "#f8fdf9", fontFamily: "'Segoe UI', sans-se
  }}>
    {/* Header */}
    <div style={{
      background: "linear-gradient(135deg, #1a6b3a, #2e7d32)", color: "white",
      padding: "20px 24px", position: "sticky", top: 0, zIndex: 100,
      boxShadow: "0 4px 20px #1a6b3a44"
    }}>
      <div style={{ maxWidth: 720, margin: "0 auto", display: "flex", alignItem
        <div style={{ fontSize: 36 }}>🥗</div>
        <div>
          <div style={{ fontWeight: 900, fontSize: 20, letterSpacing: -0.5 }}>N
          <div style={{ opacity: 0.85, fontSize: 13 }}>
            {profile.name ? `${profile.name}'s` : "Your"} personalized {profile
          </div>
        </div>
        <div style={{ marginLeft: "auto", display: "flex", gap: 10 }}>
          <button onClick={() => setStep("chat")} style={{
            background: "rgba(255,255,255,0.2)", border: "1px solid rgba(255,25
            color: "white", borderRadius: 12, padding: "8px 16px", cursor: "poi
            fontSize: 13, fontWeight: 600
          }}>💬 AI Chat</button>
          <button onClick={() => setStep("onboard")} style={{
            background: "rgba(255,255,255,0.2)", border: "1px solid rgba(255,25
            color: "white", borderRadius: 12, padding: "8px 16px", cursor: "poi
```

```
        fontSize: 13, fontWeight: 600
      }}>↩ Profile</button>
    </div>
  </div>
</div>

<div style={{ maxWidth: 720, margin: "0 auto", padding: "24px 20px" }}>
  {/* Stats */}
  <div style={{ display: "grid", gridTemplateColumns: "repeat(4, 1fr)", gap
    {[
      ["🔥", "Total Calories", `${totalNutrients.cal}`, "kcal"],
      ["💪", "Protein", `${totalNutrients.protein}g`, "daily"],
      ["⚡", "Carbs", `${totalNutrients.carbs}g`, "daily"],
      ["🫙", "Fat", `${totalNutrients.fat}g`, "daily"],
    ].map(([icon, label, val, unit]) => (
      <div key={label} style={{
        background: "white", borderRadius: 16, padding: "14px 12px",
        textAlign: "center", boxShadow: "0 2px 12px #1a6b3a11", border: "1p
      }}>
        <div style={{ fontSize: 24 }}>{icon}</div>
        <div style={{ fontWeight: 900, fontSize: 18, color: "#1a6b3a", marg
        <div style={{ fontSize: 10, color: "#888", fontWeight: 600, textTra
      </div>
    ))}
  </div>

  {/* Nutrient Progress */}
  <div style={{
    background: "white", borderRadius: 18, padding: "20px 22px", marginBott
    boxShadow: "0 2px 12px #1a6b3a11", border: "1px solid #e8f5e9"
  }}>
    <div style={{ fontWeight: 800, fontSize: 14, color: "#1a2e1a", marginBo
      📊 Nutritional Coverage
    </div>
    <NutrientBar label="Protein" value={totalNutrients.protein} max={profil
    <NutrientBar label="Carbohydrates" value={totalNutrients.carbs} max={30
    <NutrientBar label="Fat" value={totalNutrients.fat} max={100} color="#3
  </div>

  {/* AI Insights */}
  <AIInsightsPanel />

  {/* Meals */}
  {[
    ["🌅 Breakfast", [plan.breakfast]],
    ["☀️ Lunch", [plan.lunch]],
    ["🌙 Dinner", [plan.dinner]],
```

```
        ["🍎 Snacks", [plan.snack1, plan.snack2].filter(Boolean)],
      ].map(([title, items]) => (
        <MealSection key={title} title={title} icon={title.split(" ")[0]}
          items={items} onRate={handleRating} />
      ))}

      {Object.keys(ratings).length > 0 && (
        <div style={{
          background: "#e8f5e9", borderRadius: 14, padding: "14px 18px",
          border: "1px solid #c8e6c9", marginTop: 8
        }}>
          <div style={{ fontWeight: 700, color: "#1a6b3a", fontSize: 13 }}>
            🤖 ML Feedback Captured — Model will adapt your next plan!
          </div>
          <div style={{ fontSize: 12, color: "#555", marginTop: 4 }}>
            {Object.entries(ratings).map(([f, v]) => `${f}: ${v > 0 ? "👍" : "👎
          </div>
        </div>
      )}

      <button onClick={generatePlan} style={{
        width: "100%", marginTop: 24, padding: "14px", borderRadius: 14, border
        background: "linear-gradient(135deg, #1a6b3a, #2e7d32)", color: "white"
        fontSize: 15, fontWeight: 800, cursor: "pointer", boxShadow: "0 6px 20p
      }}>
        🔄 Regenerate Plan with ML
      </button>
    </div>
  </div>
);

// ─── CHAT VIEW ──────────────────────────────────────────────
if (step === "chat") return (
  <div style={{
    minHeight: "100vh", background: "#f8fdf9", fontFamily: "'Segoe UI', sans-se
    display: "flex", flexDirection: "column"
  }}>
    <div style={{
      background: "linear-gradient(135deg, #1a6b3a, #2e7d32)", color: "white",
      padding: "18px 24px", display: "flex", alignItems: "center", gap: 14,
      boxShadow: "0 4px 20px #1a6b3a44"
    }}>
      <div style={{ fontSize: 34 }}>🤖</div>
      <div>
        <div style={{ fontWeight: 900, fontSize: 18 }}>NutriAI Dietitian</div>
        <div style={{ fontSize: 12, opacity: 0.85 }}>Powered by Claude · AIML-e
      </div>
```

```
    <button onClick={() => setStep("plan")} style={{
      marginLeft: "auto", background: "rgba(255,255,255,0.2)", border: "1px s
      color: "white", borderRadius: 12, padding: "8px 16px", cursor: "pointer
    }}>← Meal Plan</button>
  </div>

  {chatHistory.length === 0 && (
    <div style={{ padding: "24px 24px 0" }}>
      <div style={{ background: "white", borderRadius: 16, padding: "18px 20p
        <div style={{ fontSize: 24, marginBottom: 8 }}>👋</div>
        <div style={{ fontWeight: 700, color: "#1a2e1a", marginBottom: 4 }}>H
        <div style={{ color: "#555", fontSize: 14, lineHeight: 1.6 }}>
          Ask me anything about nutrition, your meal plan, food substitutions
        </div>
        <div style={{ display: "flex", flexWrap: "wrap", gap: 8, marginTop: 1
          {["What should I eat before workout?", "Why is protein important fo
            <button key={q} onClick={() => { setChatInput(q); }}
              style={{
                background: "#e8f5e9", color: "#1a6b3a", border: "none", bord
                padding: "8px 14px", fontSize: 12, fontWeight: 600, cursor: "
              }}>{q}</button>
          ))}
        </div>
      </div>
    </div>
  )}

  <div ref={chatRef} style={{ flex: 1, overflowY: "auto", padding: "20px 24px
    {chatHistory.map((msg, i) => (
      <div key={i} style={{
        display: "flex", justifyContent: msg.role === "user" ? "flex-end" : "
        marginBottom: 14
      }}>
        {msg.role === "assistant" && <div style={{ fontSize: 28, marginRight:
        <div style={{
          maxWidth: "75%", padding: "12px 16px", borderRadius: 16,
          background: msg.role === "user" ? "linear-gradient(135deg, #1a6b3a,
          color: msg.role === "user" ? "white" : "#1a2e1a",
          fontSize: 14, lineHeight: 1.6, boxShadow: "0 2px 12px #00000015",
          border: msg.role === "assistant" ? "1px solid #e8f5e9" : "none",
          borderBottomRightRadius: msg.role === "user" ? 4 : 16,
          borderBottomLeftRadius: msg.role === "assistant" ? 4 : 16,
        }}>{msg.content}</div>
      </div>
    ))}
    {chatLoading && (
      <div style={{ display: "flex", alignItems: "center", gap: 10, marginBot
```

```
          <div style={{ fontSize: 28 }}>🤖</div>
          <div style={{
            background: "white", borderRadius: 16, borderBottomLeftRadius: 4,
            padding: "12px 20px", border: "1px solid #e8f5e9", boxShadow: "0 2p
          }}>
            <div style={{ display: "flex", gap: 6 }}>
              {[0, 1, 2].map(i => (
                <div key={i} style={{
                  width: 8, height: 8, borderRadius: "50%", background: "#1a6b3
                  animation: `bounce 1.2s ${i * 0.2}s infinite`,
                }} />
              ))}
              <style>{`@keyframes bounce { 0%,80%,100%{transform:translateY(0)}
            </div>
          </div>
        </div>
      )}
    </div>

    <div style={{
      padding: "16px 24px 24px", background: "white", borderTop: "1px solid #e8
      boxShadow: "0 -4px 20px #1a6b3a11"
    }}>
      <div style={{ display: "flex", gap: 12 }}>
        <input
          value={chatInput}
          onChange={e => setChatInput(e.target.value)}
          onKeyDown={e => e.key === "Enter" && sendChat()}
          placeholder="Ask your AI dietitian anything..."
          style={{
            flex: 1, borderRadius: 14, border: "1.5px solid #e0e0e0", padding:
            fontSize: 14, outline: "none", fontFamily: "inherit", transition: "
          }}
          onFocus={e => e.target.style.borderColor = "#1a6b3a"}
          onBlur={e => e.target.style.borderColor = "#e0e0e0"}
        />
        <button onClick={sendChat} disabled={chatLoading || !chatInput.trim()}
          style={{
            background: "linear-gradient(135deg, #1a6b3a, #2e7d32)", color: "wh
            border: "none", borderRadius: 14, padding: "12px 20px", cursor: "po
            fontSize: 20, boxShadow: "0 4px 14px #1a6b3a33", transition: "all 0
          }}>➤</button>
      </div>
    </div>
  </div>
);
```

```
  return null;
}
```